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Faisal D'Souza, NCO
Office of Science and Technology Policy
Executive Office of the President
2415 Eisenhower Avenue
Alexandria, VA 22314

Submitted by email to [REDACTED]

Re: Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan ("Plan")

I. Introduction

Oklo strongly supports the Office of Science and Technology Policy's (OSTP) focus on policy actions required to support U.S. global leadership in artificial intelligence (AI) innovation. Advanced nuclear technology is well positioned to provide 24-7 reliable, affordable and zero-emission energy to meet the vast electricity needs to power AI. This submission describes specific actions the Trump Administration can take or support to accelerate nuclear adoption and strengthen domestic AI infrastructure.

AI leadership is a civilization-level challenge, and we face a geopolitical imperative to achieve AI supremacy. Energy is the foundation upon which America's AI future depends. AI supremacy requires and will create an unprecedented strain on our ability to deliver new energy generation. Coupled with the reindustrialization of America, delivering the infrastructure for AI dominance runs entirely counter to recent history and trends. We will need an "all-of-the-above" approach just to manage our base case for AI needs, yet, at the same time, we have the opportunity to fundamentally reshape the global balance of power in energy dominance, the necessary condition for AI dominance.

To address America's needs for new energy, our recommendations can be separated into three categories: (1) secure a domestic fuel supply for nuclear reactors; (2) encourage commercial recycling of spent nuclear fuel; and (3) create a whole-of-government approach to nuclear acceleration. The pan-government approach to energy dominance should include comprehensive deregulation through the NRC, first of a kind financing for critical projects, comprehensive Congressional reform of nuclear's regulatory regime, and a coordinated effort across our national security apparatus to ensure AI supremacy.

Given the urgent need to address skyrocketing electricity demand due to AI and advanced computing, we urge the Administration to incorporate our recommendations in the AI Action Plan and make clear that advanced nuclear power is a critical component of America's ability to "sustain and enhance America's global AI dominance^[1]."

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II. About Oklo

Oklo is developing fast fission power plants to provide clean, reliable, and affordable energy at scale as an independent power producer. We are at the forefront of the efforts to transform the technological basis and business model associated with nuclear power in America. Building upon a half century of proven technology, Oklo hopes to create a SpaceX moment for advanced nuclear, unleashing its potential to deliver American energy dominance, to power our AI revolution, and to ensure, long term, that our post-AI future in America is not post-industrial.

Our Technology

Oklo's reactors are based on proven liquid metal-cooled sodium fast reactor technology, with over 400 reactor-years of operational experience worldwide. Our reactors are inherently safe, requiring no operators, pumps, or mechanical systems to remain cool. We are building upon America's investment in cutting edge nuclear technology in the first Atomic Age.

In particular, using sodium instead of water as a coolant provides key advantages:

- No high-pressure containment – Traditional water-cooled reactors require massive civil infrastructure (concrete and steel) to withstand high-pressure systems, driving up costs.
- Lower capital costs – Oklo's non-pressurized design eliminates these requirements, significantly reducing construction costs.
- Faster deployment and scalability – 70% of Oklo's reactor components come from existing industrial supply chains, making them far more cost-efficient and scalable than traditional nuclear-specific components.

Oklo's ability to recycle used nuclear fuel, and our use of fast reactor technology that can utilize it, also gives us a significant cost advantage over developers whose reactors rely solely on fresh fuel. By leveraging recycled fuel, Oklo expects to cut fuel costs by up to 80%, making our solution more cost-effective than both conventional and advanced nuclear technologies that rely on fresh fuel. Oklo's groundbreaking approach is redefining nuclear energy—making it safer, faster to deploy, and more cost effective than ever before, at a time when access to clean baseload energy is a civilizational priority.

A New Model for Providing Power

More significantly, Oklo, and the cohort of advanced nuclear companies that have emerged in recent years, represent a wholesale rethinking of how to leverage the benefits of mature nuclear power generation in different form factors and fundamentally different business models than the current approach.

Under Oklo's build-own-operate model, customers must neither provide capital commitment nor bear most project risks. They can simply purchase the energy product they need. This model also increases the efficiency of our deployments by improving access to financing, encouraging alignment of incentives (e.g., between facility design and licensing), and fostering a culture of learning and growth across the design, licensing, construction, and operation of our



powerhouses.

Our business model eliminates the significant regulatory cost and risk burden on the customer that has historically been typical in the nuclear power industry. To date, the advanced fission industry is largely following the blueprint in which reactor developers seek U.S. Nuclear Regulatory Commission (NRC) approval of their reactor design and seek to sell that design to utility customers who will bear the future burden of licensing, building, and operating a power plant using that design.

In this model, the reactor developer is not incentivized to consider the full lifecycle costs of the design and licensing decisions they make in gaining a design approval, and they may lock in higher costs by obtaining approval of a less-than-optimal design. Oklo's model corrects this inefficiency, as we are responsible for licensing and operating the reactors we design, leading to a relentless focus on safety- and security-by-design. Lifecycle regulatory, construction, operating, maintenance, and decommissioning costs are then built into our contract prices, meaning there are no surprise costs for the customer. The market has rewarded this approach with a more than 14GW pipeline of commitments from prospective customers, most recently in an agreement with cutting-edge data provider Switch, spanning twenty years.

Oklo received a site use permit from the U.S. Department of Energy, was awarded fuel material from Idaho National Laboratory, submitted the first advanced fission custom combined license application to the U.S. Nuclear Regulatory Commission, and is developing advanced fuel recycling technologies in collaboration with the U.S. Department of Energy and U.S. National Laboratories.

III. Policy Recommendations

Advanced nuclear reactors will play a critical role in ensuring that the U.S. can unilaterally meet the power needs of AI technology and maintain a reliable grid with affordable energy. Electricity demand is skyrocketing worldwide. In December 2023, the North American Electric Reliability Corporation more than doubled its 9-year electricity demand forecast from the prior year, from approximately 220 to approximately 560 gigawatt hours of growth.^[1] This is due in large part to high performance computing and artificial intelligence applications at data centers, manufacturing and industrial growth and electrification. Although data centers currently account for just 1 percent of electricity usage globally, this percentage is expected to drastically increase to meet demand.^[2] This has already occurred in some areas – in the Commonwealth of Virginia the share is already more than 25 percent.

Solving these power challenges is a national and economic security imperative. If our domestic energy supply is insufficient to meet AI infrastructure needs in the U.S., AI developers will likely turn to other countries with affordable and abundant energy sources, some of which may be adversaries of the U.S. A strong AI infrastructure will attract and maintain AI developers, to ensure that AI technology aligns with U.S. interests and values.

This surge in demand faces a stark challenge; scaling existing methods of power cannot deliver what America needs. Our country will need an all-of-the-above approach to supply energy in the



short to medium term. But we cannot let short-term urgency not overshadow unlocking energy dominance for decades to come. The existing nuclear sector is an affordable, consistent, reliable and zero-emission energy source, qualities critical to the compute sector and to U.S. competitiveness. And it represents a small fraction of what nuclear can supply for America's energy needs.

Advanced nuclear energy is uniquely well positioned to address America's energy reliability and security challenges for AI and our industrial base more broadly. Advanced nuclear reactors share the benefits of large light water reactors, without the liabilities that have capped our ability to deploy gigawatt scale nuclear over the past 30 years. Benefits of advanced nuclear, including a smaller footprint and modularity, make it a good fit for AI energy needs and require commensurate investment in scaling across the advanced nuclear supply chain, including workforce and fuel supply.

Additionally, enabling policies for efficient licensing and spent fuel utilization will support cost effective commercialization. The regulatory approval process for nuclear energy in the U.S. largely falls to the Nuclear Regulatory Commission (NRC), but other agencies also have environmental permitting roles and the U.S. Department of Energy (DOE) is central to policies relevant to research and development activities, as well as a national security apparatus that can support civil nuclear energy development. Examining and, if necessary, reforming the policy roles of each of these agencies is essential as the Trump Administration pursues efforts to enable nuclear adoption.

To further accelerate and support commercialization of advanced nuclear reactors, the following policy changes are needed to allow nuclear power to serve as a critical energy resource for data centers and AI innovation.

Secure a Domestic Fuel Supply

Advanced nuclear is poised to deploy rapidly in the U.S. with advancements in next generation reactors, but if we are to be successful, we must think beyond just reactor designs and think holistically about the supply chain, and in particular advanced reactor fuel supply. The U.S. has relinquished its leadership in fuel production over the past 30 years. Currently, Russia dominates the global nuclear fuel supply chain. The invasion of Ukraine and subsequent ban on Russian enriched uranium imports stressed the need for nuclear fuel security, which expedited many programs currently in process.

Accelerate HALEU Availability

Hyperscalers and AI companies have publicly made not just significant commitments to nuclear in general, but have launched deep partnerships with advanced nuclear companies (e.g., Switch and Oklo, Google and Kairos, Amazon and X-Energy, among others). A common thread between these three relationships and almost all non-water cooled advanced nuclear reactors is the need for high assay low enriched uranium (HALEU) which is currently not commercially available. The Nuclear Fuel Security Act required the DOE to establish a High Assay Low Enriched Uranium (HALEU) supply chain and develop additional domestic low-enriched uranium (LEU) enrichment capabilities for the light water reactor fleet. DOE made awards in



late 2024 for HALEU and LEU enrichment and deconversion. The AI Action plan should encourage DOE to:

- execute the associated contracts and instruct the companies to begin work licensing and deploying infrastructure immediately;
- ensure commercial production of HALEU in metal and oxide form starts in 2028 at the latest;
- review and amend planned contract language and milestones to enable speedy development;
- quickly identify the top enrichment and deconversion companies capable of meeting the desired timeline and ensure they are adequately supported to reach commercialization quickly, either through funding or removing administrative barriers;
- finalize and publish the mechanism and cost structure that it will use to sell the HALEU it acquires back to the industry; and
- ensure appropriate balance between support for expansion of LEU capacity and buildout of new HALEU capacity.

Unlock Sources of New Material

Independent of fresh HALEU, the federal government has considerable stockpiles of material, including spent research reactor fuel, that could be processed into HALEU or replacement fuels for HALEU. The White House and the AI Action plan should include instructing DOE to develop and implement programs to process various DOE materials (including spent research reactor fuel) to produce HALEU. This effort should recognize that advanced reactor technologies differ in the purity of fuel they require; some designs, including Oklo, include reactors capable of utilizing fuel with significantly higher impurity levels than typical “fresh HALEU.” The robustness and flexibility of these designs dramatically expands the pool of material that can be transformed from a liability into an asset and be utilized as a fuel source. DOE should not rule out any plausible source of material without first consulting the industry on its potential use in their reactors.

The DOE and the National Nuclear Security Administration (NNSA) must work together in a unified effort, and it is recommended that DOE and NNSA work closely with industry as these new programs are implemented to ensure that the needs of industry are met in a timely manner. These programs must move at the pace of industry to ensure that advanced nuclear can meet the power demands of AI on their schedules.

Encourage and Incentivize Nuclear Fuel Recycling

Spent nuclear fuel in the U.S. is currently stored at every reactor site and statutorily is slated to eventually go to a deep geologic repository. Legal challenges, community concerns and statutory limitations have stymied long term disposal solutions for decades. Thanks to U.S. innovation, spent nuclear fuel can be recycled by commercial companies and used to power advanced



reactors while reducing the cost of fuel and the burden on the fuel supply chain. It is being done at our national labs today. Idaho National Laboratory is processing spent fuel from Experimental Breeder Reactor II into HALEU. Traditional light-water reactors cannot make ready use of recycled fuel. However, fast spectrum reactors, like Oklo's, offer a unique solution to reduce operating costs, shrink waste volumes from the existing conventional reactor fleet and U.S. surplus materials (which currently impose costs on taxpayers to manage and dispose of), and recover latent nuclear energy in these materials that can provide 24/7 generation capacity. Even more significantly, over the next ten years, industrialization of recycling could reduce fuel costs by as much as 80% for advanced reactors that can leverage recycled fuel. Recycling will not fully replace the need for a domestic supply chain of HALEU, but it will enable accelerated deployment of advanced reactors to support a rapid buildout of AI capability.

As an added benefit, recycling will begin to address the increasing inventory of commercial spent nuclear fuel. With the capacity to reduce the total volume of spent nuclear fuel by more than 80%, recycling represents an opportunity to turn one of our country's most intractable problems (spent nuclear fuel) into an asset that ensures permanent energy superiority. In addition, recycling may be the quickest way to begin to reduce the annual taxpayer liabilities, on the order of \$700 million a year, due to the inability of DOE to pick-up spent nuclear fuel from commercial reactors beginning in 1998.

The White House policy on commercial recycling of spent nuclear fuel has been, at best, inconsistent over the years. The Biden Administration's policy, derived from National Security Memorandum 19, is that it does not encourage commercial recycling but does support continued research and development. The antiquated opposition to recycling within the nonproliferation community and within some areas of DOE leans on old arguments that the U.S. should lead by example by not recycling, which do not reflect the best practice demonstrated in the rest of the world. Many countries, including France, Russia, and Japan, currently recycle or plan to recycle spent nuclear fuel, without creating issues that implicate non-proliferation concerns.

The AI Action plan should include issuing a policy statement from the White House that expresses support for commercial recycling of spent nuclear fuel that does not separate out a pure stream of plutonium. This stance aligns with national energy objectives, strengthens nonproliferation efforts, enhances homeland security, and fosters technological innovation while improving government efficiency and reducing costs for taxpayers.

Create a Whole-of-Government Approach to Nuclear Acceleration

Unlocking nuclear energy's potential to help meet AI energy infrastructure's growing demand requires a whole-of-government approach. Initiatives across the government are already underway to advance nuclear energy through regulatory agencies and Congress. The White House should coordinate efforts with them to leverage resources, investments and authority to advance nuclear energy, particularly with respect to powering data centers. The AI Action Plan should address the following:

Deregulation: The Nuclear Regulatory Commission (NRC) has the tools to accelerate deployment of nuclear energy. The NRC is an independent agency responsible for



administering reactor licensing and renewal, and conducting oversight of such facilities, and regulating the management of spent fuel. Importantly, the NRC is in the midst of implementing the ADVANCE Act of 2024^[3], a bill signed into law in July 2024 that created new reforms for the agency.^[4]

But there is more to be done, including streamlining the design basis threats (DBT) requirements for security of nuclear material and power operations. The NRC currently applies a single Design Basis Threat (DBT) to all commercial nuclear power facilities, based on its responsibility under the Atomic Energy Act to consider a wide array of adversary capabilities and its applicability to facilities. While this approach historically aligned with large-scale nuclear plants that posed significant offsite risks, emerging reactor technologies—such as small, passively safe reactors designed for decentralized power generation, including AI computing infrastructure—do not present the same threat profile. A clarification of NRC policy to establish a graded DBT framework, calibrated to power output, would accelerate nuclear deployments without compromising safety and maintain security effectiveness in line with existing legislation.

Similarly, to the extent possible, we encourage the White House to work with the Federal Energy Regulatory Commission (FERC) on policies and dockets that can accelerate bringing new resources online through faster interconnection review and addressing resource adequacy. FERC has several areas through its policies where it can help get energy resources online faster, and we would encourage them to prioritize those. We also encourage the White House and all other relevant agencies including DOE to streamline the permitting process for nuclear power facilities, including by working with state and local governments to do the same. OSTP, the Office of Management and Budget, and DOE should collaborate to make budgetary decisions that support advanced nuclear acceleration and data centers.

Financing First of a Kind: The White House should work with DOE, DOD and other agencies to effectuate acceleration and support commercialization of advanced nuclear reactors. The federal government has tremendous tools to leverage that would support first of a kind to nth of a kind deployments of advanced reactors, especially those that support data centers. Specifically, the White House and DOE should create a public plan to make eligible new energy sources that support data centers in existing DOE financing programs. The White House should extend these financing tools to fuel supply and recycling facilities necessary for advanced reactor deployment.

Congress: The White House should also work with Congress to support additional reforms in an ADVANCE Act 2.0 proposal. Reforms could include ideas such as eliminating the required uncontested mandatory hearings, modernizing the Advisory Committee on Reactor Safeguards, and providing clarity on licensing recycling facilities via the Part 70 licensing pathway. In addition, U.S. policy should permit the NRC to apply a lower amount of regulation for commercial licensing of reactors less than 100 Mwe, similar to NRC's process for licensing medical therapy and research reactors.



National Defense/Security: Advanced nuclear reactors can also contribute to the U.S. government’s AI infrastructure needs. The White House should work with the Department of Defense and other relevant agencies, such as the NNSA and national laboratories, to identify AI infrastructure needs specifically for government use. AI infrastructure includes the hardware required to develop AI technologies, data centers and energy sources.

The White House should consider innovative ways to make such investments, including consulting private sector stakeholders, to ensure the federal government has access to sufficient AI capabilities.

IV. Conclusion

Advanced nuclear reactors must be part of the energy solution to ensure U.S. leadership in AI. The technology offers an affordable, consistent, reliable and zero-emission energy source, which are characteristics critical to the compute sector and to U.S. competitiveness. Oklo’s recommendations above will help realize advanced nuclear’s benefits as quickly as possible. Failing to address these nuclear energy challenges, and the energy supply shortage more broadly, will put America’s AI goals and national security at risk.

[1] North American Electric Reliability Corporation. (2023), 2023 Long-Term Reliability Assessment, https://www.nerc.com/pa/RAPA/ra/Reliability%20Assessments%20DL/NERC_LTRA_2023.pdf.

[2] Between 2024 and 2030, electricity demand for data centers in the United States is expected to increase by about 400 terawatt-hours at a CAGR of about 23 percent. McKinsey & Company, How data centers and the energy sector can sate AI’s hunger for power (September 17, 2024), <https://www.mckinsey.com/industries/private-capital/our-insights/how-data-centers-and-the-energy-sector-can-sate-ais-hunger-for-power>.

[3] Nuclear Energy Leadership Act, Energy Act of 2020, ADVANCE Act, Nuclear Fuel Security Act.

[4] P.L. 118–67.

[1] Removing Barriers to American Leadership in Artificial Intelligence, Executive Order (January 23, 2025), <https://www.whitehouse.gov/presidential-actions/2025/01/removing-barriers-to-american-leadership-in-artificial-intelligence/>.

